HW4 113705015 [illegible]

C105061Q14

14 $\frac{\max(s_1, s_2)}{\min(s_1, s_2)} = \frac{128.2^2}{15.6^2} = 3.628 > 3$, significance difference

① $H_0: \sigma_1^2 = \sigma_2^2$ vs $H_a: \sigma_2^2 > \sigma_1^2$

② $\alpha = 0.05$

③ Test statistic:

$F = \frac{S_2^2}{S_1^2} \sim F_{29,29;0.05}$

④ $F^* = \frac{(28.2)^2}{(15.6)^2} = 3.268$

⑤ p-value $= P(F > 3.268) \approx 0.001 < F_{29,29;0.005} = 2.66$

must be, cause $3.268 > 2.66$

Because $P < 0.005$, so $\rightarrow$ $P < 0.05$ (α)

$\hookrightarrow 0.001 < 0.05$

⑥. ∴ Reject H_0, there's sufficient evidence to prove $\sigma_2^2 > \sigma_1^2$

C105061Q16

16-① $H_0: \sigma_1^2 = \sigma_2^2$ vs $H_a: \sigma_1^2 \neq \sigma_2^2$

② $\alpha = 0.05$

③ Test statistic:

$F = \frac{S_1^2}{S_2^2} \sim F_{14,14;0.025}$

④ $F^* = \frac{(114)^2}{(103)^2} = 1.225 < 2.95$

JOYKO® 36 Lines, 6 mm

No
Date

(5) p-value $= 2(F > 1.225) > 0.1$ ⇒ Because $P > 0.1$, so also $P > 0.05$
from F-table
$\approx 0.7074 > 0.05 = \alpha$

(5) F-test:

$F = 1.225$

(6) ∴ Non-reject H_0

C10S06 Q22

22. (1) $H_0: \sigma_1^2 = \sigma_2^2$ vs $\sigma_1^2 \neq \sigma_2^2$

(2) $\alpha = 0.05$

(3) Test statistic:

$$F = \frac{S_2^2}{S_1^2} \sim F_{14,14;0.025}$$

(4) $$F^* = \frac{2.96}{1.54} = 1.9221$$

(5) p-value $= 2P(F > 1.9221) > 0.05$

$\approx 0.2 > 0.05$

(6) ∴ Non-reject H_0

b. 99% CI $= \left(\frac{S_1^2}{S_2^2} \cdot \frac{1}{F_{(n_2-1),(n_1-1);0.005}} , \frac{S_1^2}{S_2^2} \cdot F_{(n_1-1),(n_2-1);0.005} \right)$

$= \left(\frac{2.96}{1.54} \cdot \frac{1}{4.25} , \frac{2.96}{1.54} \cdot 4.25 \right)$

(using $df_1 = 15$ (C10S02))

$= (0.452 , 8.169)$

C10S07 Q11

11. $\bar{x}_1 = 1.875$ $\bar{x}_2 = 2.625$

$S_1^2 = 0.696$ $S_2^2 = 0.839$

$\frac{S_2^2}{S_1^2} = \frac{0.839}{0.696} \not> 3$, so same variances

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{df} = \frac{7 \times 0.696 + 7 \times 0.839}{14} = 0.768$$

a. (1) $H_0: \mu_1 = \mu_2$ vs $H_a: \mu_1 \neq \mu_2$

(2) $\alpha = 0.05$

(3) Test statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\quad}} \sim t_{14,0.025}$$

No
Date

(4) $t^* = \dfrac{(1.875 - 2.625)}{\sqrt{\frac{0.768}{8} + \frac{0.768}{8}}} = -1.712 > -t_{14, 0.025} = -2.145$

(5) R.R: $\{t \mid t < -t_{14, 0.025} = -2.145$ or $t > 2.145\}$

(6) non-reject H_0

CIUS07 Q12

12.

Person	Stimulus 1	Stimulus 2	D_i
1	3	4	-1
2	1	2	-1
3	1	3	-2
4	2	1	1
5	1	2	-1
6	2	3	-1
7	3	3	0
8	2	-3	-1

$\bar{D}_i = -0.75$

$S_D^2 = 0.7857$

Define $D_i = u_1 - u_2$

(1) $H_0: u_D = 0$ vs $H_a: u_D \neq 0$

(2) $\alpha = 0.05$

(3) Test statistic:

$$t = \frac{\bar{D}_i - u_D}{S_D / \sqrt{n}} \sim t_{7, 0.025}$$

(4) $t^* = \dfrac{-0.75}{\sqrt{\frac{0.7857}{8}}} = -2.393 < -t_{7, 0.025} = -2.365$

(5) R.R: $\{t \mid t < -t_{7, 0.025} = -2.365$ or $t > 2.365\}$

(6) $\therefore$ Reject H_0